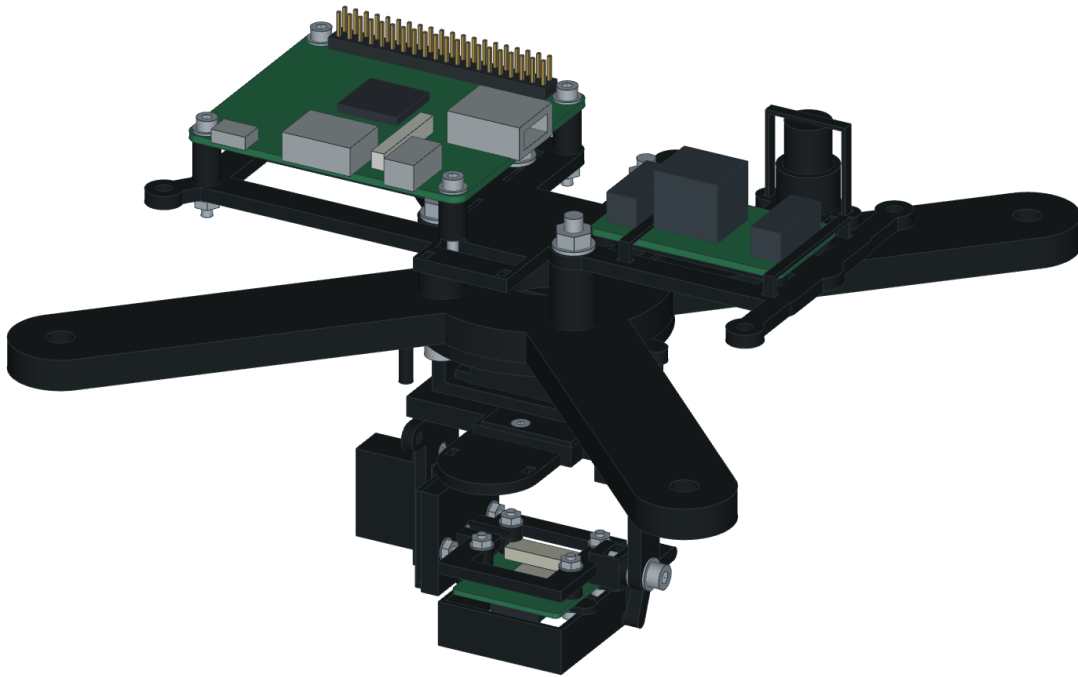


Build the payload

Pi 3A+ / Camera Module 3 Standard / two micro servos. Dry bench alternative; deck/yoke r0.1.1, other prints as registered.



THE MISSING PRINTED STRUCTURE IS NOW INCLUDED

A fixed electronics deck, both servo mounts, stock-horn retainers, supported camera cradle, removable pivot support, optical hood and electronics cover. Every illustration uses the supplied STL files.

Fit before the full print

Servo and power-module dimensions remain provisional. Start with the servo coupon and compare the dimension table on page 3 with your received parts. These models are not physically tested.

CAD checks cover rigid geometry, fasteners and motion. Suspended operation still needs mass, strength, balance, cable-flex and electrical checks.

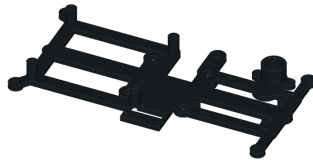
01 Printed parts

Print the listed quantity. Dark cores and white shells are fabrication meshes; electronic and metal reference meshes are not functional prints.



P01 Existing spider

1 x / r0.1.0



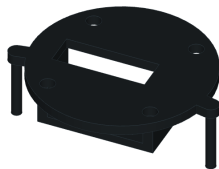
P02 Electronics deck

1 x



P03 Spider spacer

4 x / 14 mm



P04 Pan servo mount

1 x



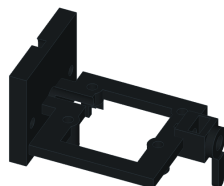
P05 Pan yoke

1 x



P06 Pivot support

1 x / removable



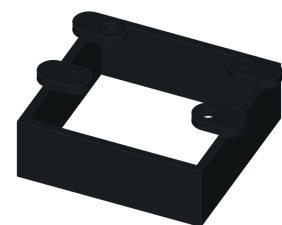
P07 Camera cradle

1 x



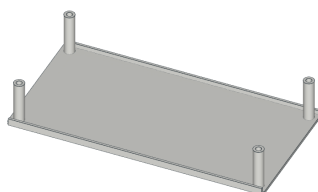
P08 Horn retainer

2 x / identical



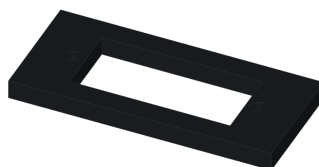
P09 Camera hood

1 x



P10 Electronics cover

1 x / removable



T01 Servo fit coupon

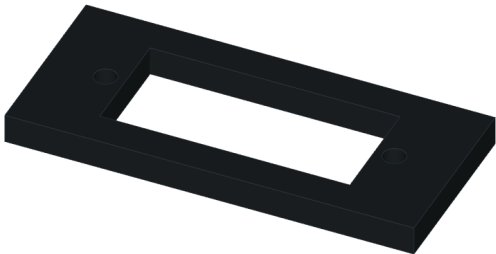
1 test print / not installed

PETG starting point:
0.2 mm layers, 3+ walls.
Solid small parts.
Supports on complex brackets.

All STLs are in mm and shifted onto Z=0. Brackets need support planning in the slicer. Remove support scars from mating faces and holes; do not enlarge the horn pocket until it becomes loose.

02 Measure before printing

The stock Pi/camera hole patterns are documented. The servo, horn, converter and capacitor must be checked against received hardware.



Test both servos
The body should pass the coupon without force; both lug holes must align. Also measure the shaft offset and height: the coupon cannot check either.

Interface	Nominal model / required check
Servo body	20 x 8.5 x 18 mm; window 20.8 x 9.3 mm
Servo ears	27 mm span; 24 mm hole pitch; 1.7 mm holes
Output shaft	5 mm off body centre; tip 22 mm above case bottom
Stock horn	8 mm hub; one 12 mm arm; hub 2 mm / arm 1.5 mm thick
Converter / capacitor	45 x 25 x 15 mm / 10 mm diameter x 16 mm can
Pi / camera holes	58 x 49 mm, M2.5 / 21 x 12.5 mm, M2

Edit hardware/lib/payload-mounts.scad and the matching reference models if the parts differ, then rebuild and repeat the checks. Use the original servo horns and centre screws; there is no printed spline.

The countersunk servo and horn-retainer screws are intentional. Proud socket heads interfere during rotation. Check the real head diameter and recess seating; no washers go under countersunk heads.

03 Install the pan servo

Do this before attaching the mount to the spider. Its large upper opening passes the servo and both mounting ears.



1 Insert from above

Feed the output shaft down through P04. The case passes the body window; the mounting ears rest on the lower lug plate.

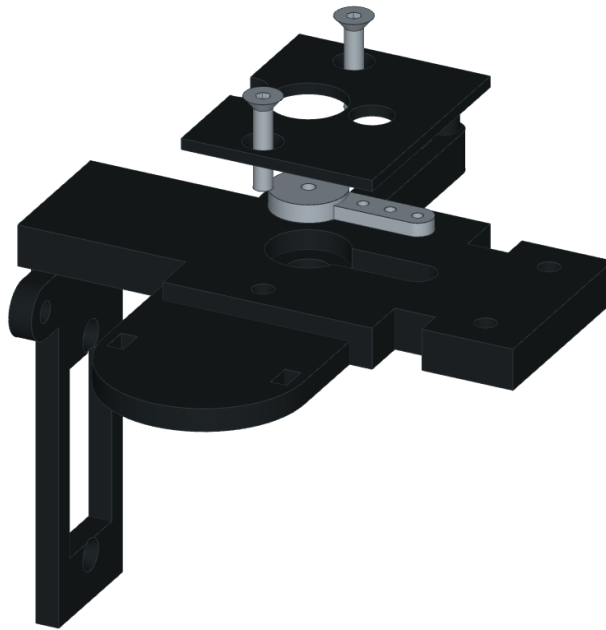
2 Use recessed ear screws

2 x M1.6 x 6 countersunk screws enter upward from below. Put the two M1.6 nuts above the servo ears. The screw heads must sit flush.

Route the servo lead through an open side with slack at the case exit. Ear holes, nut clearance and centre-screw thread are supplier-dependent; never force an oversized screw into the servo.

04 Couple the pan yoke

Set the actual servo neutral with its controller, then disconnect power before fitting the horn. Keep the horn orientation repeatable.



1 Capture the supplied horn

Seat the stock horn in the upper pocket of P05. Fit P08 over it: the underside rib bears on the horn arm. Remove free play with a fitted shim if needed.

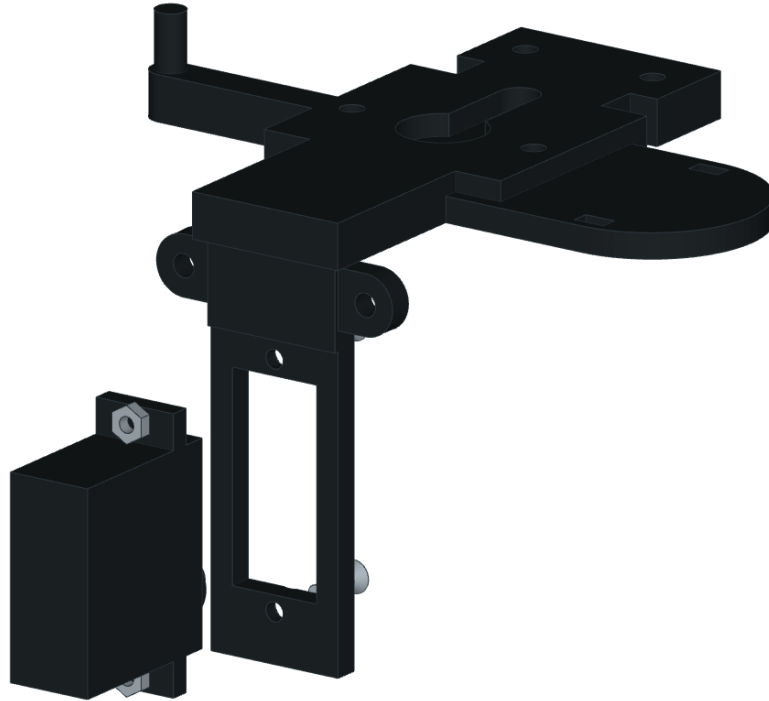
2 Retain and engage

Use 2 x M2 x 10 countersunk screws, two lower washers and two nuts. Push the captured horn onto the pan spline and fit the OEM centre screw through the access hole below.

The rib and pocket match the stated nominal horn only. Do not clamp a different horn by force. At neutral, a short driver reaches the retainer screws from the side below the pan mount; see the access-check limits.

05 Install the tilt servo

Leave the opposite pivot support off while engaging the camera horn. P06 is removable so the yoke does not need to be bent apart.



1 Insert from the outside

Slide the second servo into the left-side opening of P05. Its output shaft points inward toward the camera; the two ear holes align vertically.

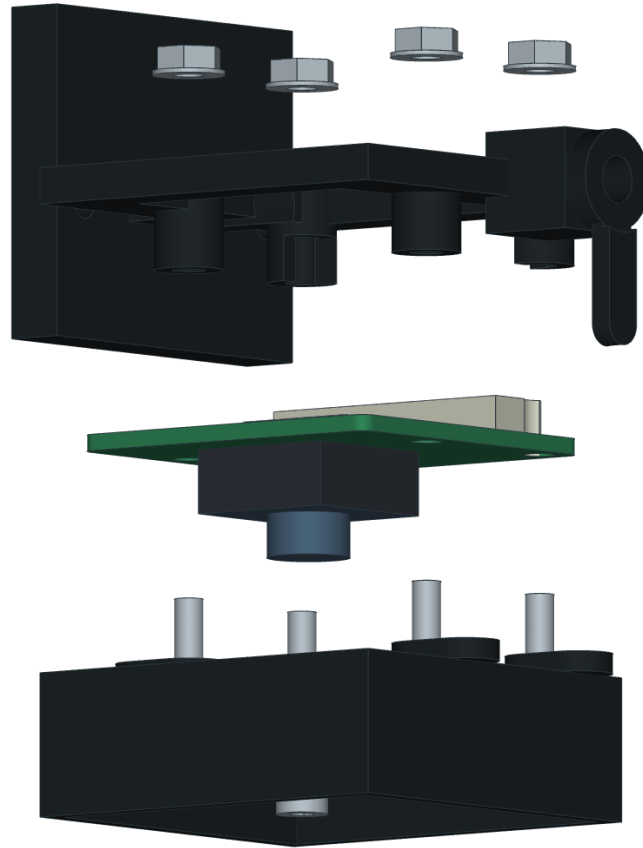
2 Recess the two screw heads

Use 2 x M1.6 x 6 countersunk screws from the inner face of the lug plate, with nuts outside the servo ears. Install these before the camera cradle blocks tool access.

At this stage both servos should be retained independently of their horns. The exposed lead exits need relaxed wire loops; the geometric references do not model the actual servo lead or plug.

06 Prepare the camera cradle

The camera mounts 5 mm off the tilt-axis centreline. This leaves access to the stock horn screw without crossing a camera mounting hole.



1 Fit the tilt horn and retainer

Capture the second stock horn in P07 with P08, 2 x M2 x 8 countersunk screws, two washers behind the carrier and two nuts. Slide the cradle onto the tilt spline while P06 is removed.

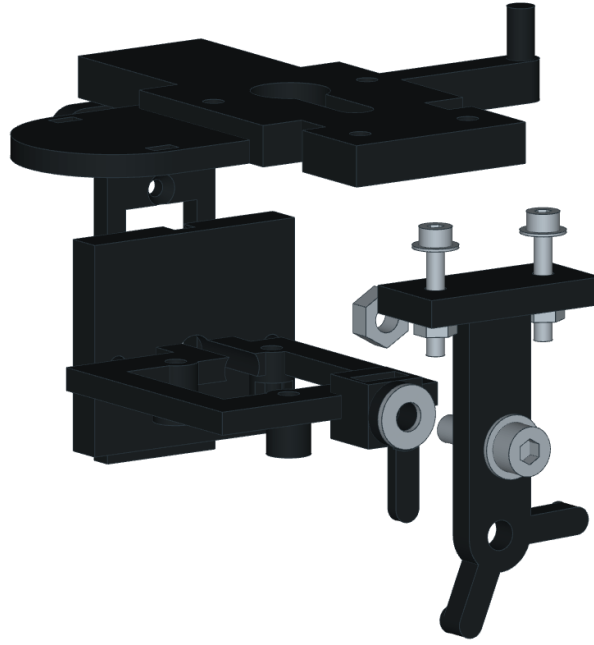
2 Secure the centre screw first

Fit the OEM centre screw from inside the cradle through its lateral access bore. Fit the camera, hood and four camera bolts after this screw is secure.

Camera stack: M2 x 12 screw + front washer > hood tab > PCB > integral cradle spacer > carrier > rear washer + nut. Use four of each. The rear CSI connector has a clearance recess.

07 Close the supported pivot

The M3 pivot provides the opposite-side support. It must rotate freely without squeezing the cradle against the yoke.



1 Drop in the pivot nut

Put the M3 plain nut into the open-top pocket in the right cradle boss. Slide P06 onto the yoke from below; its stop pins face the moving cradle tab.

2 Fasten the removable support

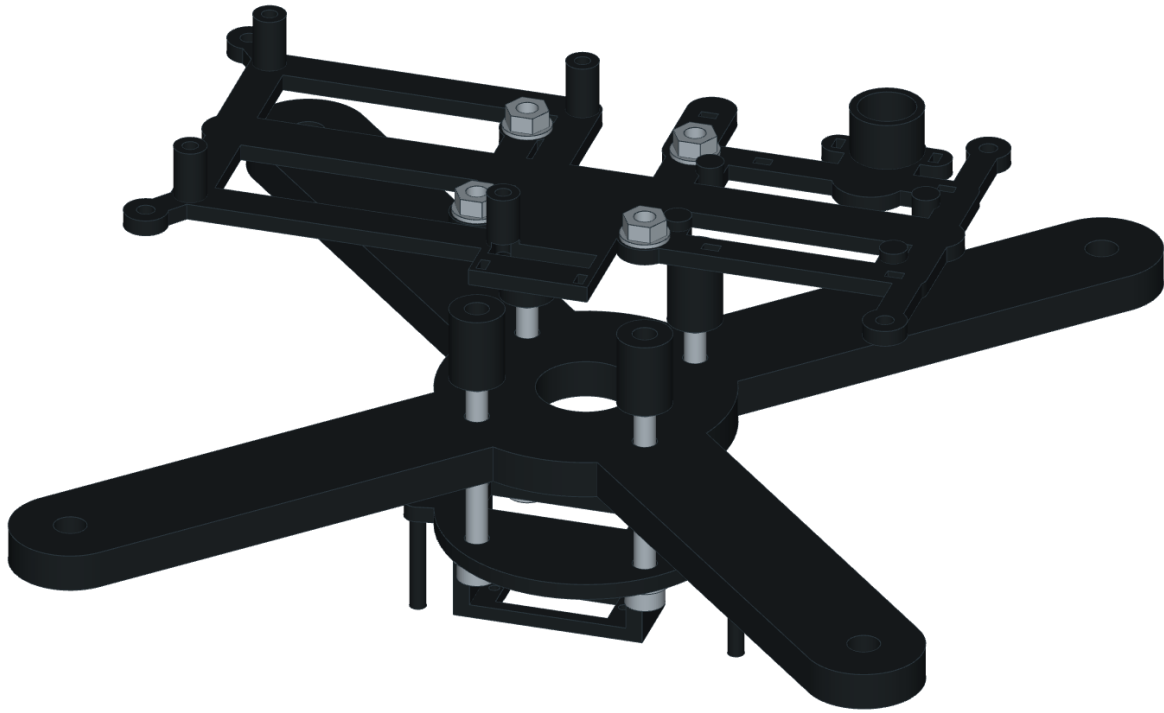
Use 2 x M2 x 12 screws, four M2 washers and two nuts through the top pad of P05 and flange of P06.

3 Fit the pivot screw and shim

Insert M3 x 12 with a 0.5 mm outer washer. Place the 0.7 mm shim in the side gap. Adjust for free travel, then retain the threaded joint without locking the pivot against the support.

08 Attach to the spider

Build the mechanical frame before fitting the Pi. This leaves the four upper frame nuts accessible.



1 Stack the three levels

P04 pan mount below the existing spider; four P03 spacers above it; P02 electronics deck on top. Align the existing 31.11 mm square mounting pattern.

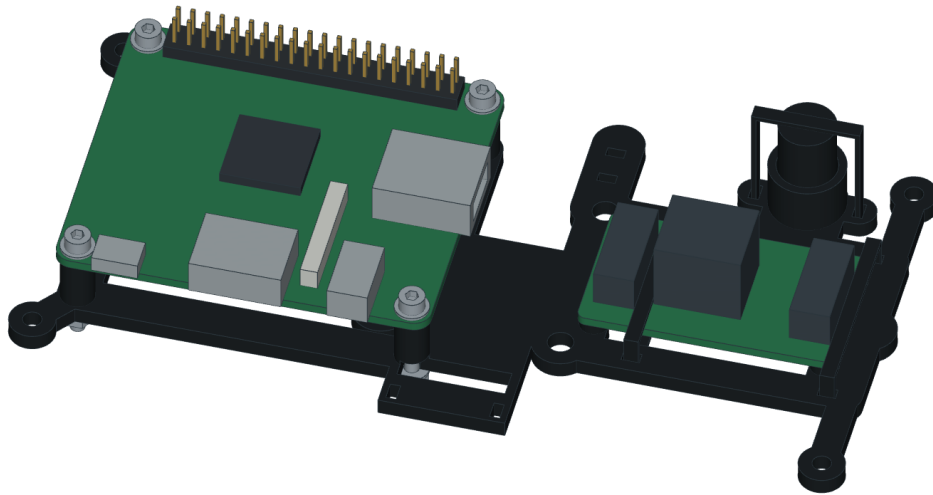
2 Fit four M4 through bolts

Insert M4 x 35 upward from below, with a washer at each end and a plain nut above the deck. Tighten evenly without crushing printed spacers.

The Pi is installed later. Its underside clears the represented frame-screw tips; replacing these bolts with longer ones requires another check.

09 Mount compute and power

All heavy electronics stay on the fixed deck. Disconnect power while mounting boards and connecting the camera ribbon.



1 Mount the Pi on four posts

Use 4 x M2.5 x 20 screws, eight M2.5 washers and four nuts. Insert the microSD card and connect the CSI cable while the cover is off.

2 Retain the converter

Seat the board on its four pads. Two 2.5 mm insulated ties pass through the deck slots and over clear PCB strips. Confirm the real module has no components or terminals under those strips.

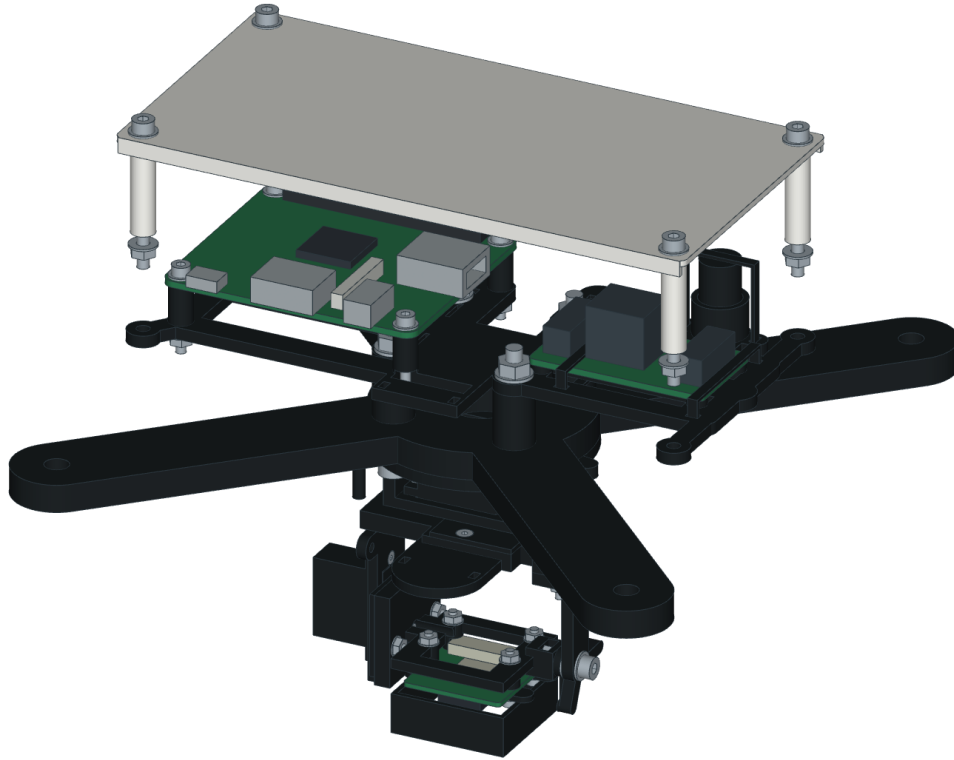
3 Fit the capacitor

Put an insulating sleeve on the leads, seat the can in its loose cup, and loop a tie over the can through the two side slots. Keep polarity visible and leads separated.

Validate regulated 5 V and polarity before connecting electronics. Servo supply comes from the regulated rail; GPIO provides signals, not motor power.

10 Route cables; add the cover

The 60 mm flat CSI mesh is only a reference segment. It is not the cable length needed by this assembled mechanism.



Provide two relaxed ribbon loops

Use the broad tie pads on the deck and pan yoke with a soft protective wrap. Route the fixed-to-pan loop through an open gap between spider arms and outside the servo sweep. Leave a separate pan-to-camera loop for tilt. Keep the CSI connector accessible.

Hand-check all four pan/tilt corner poses with the actual ribbon before choosing its length. Stop before any tension, twist concentration, sharp crease or rubbing. The CAD check does not simulate cable flex.

Fit P10 last: 4 x M3 x 35, eight M3 washers and four nuts. The cover lifts straight off after its bolts are removed. It is a splash shield; the exposed gimbal and electronics are not waterproof.

11 Hardware for one assembly

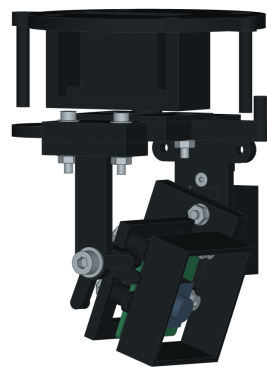
Counts below follow the illustrated assembly. Plain nuts are used in the checked envelopes. Confirm actual screw heads and supplied servo hardware.

Location	Screws	Nuts / washers
Spider / deck	4 x M4 x 35	4 M4 nuts; 8 M4 washers
Pi	4 x M2.5 x 20	4 M2.5 nuts; 8 M2.5 washers
Servo ears	4 x M1.6 x 6 COUNTERSUNK	4 M1.6 nuts; no head washers
Pan horn retainer	2 x M2 x 10 COUNTERSUNK	2 M2 nuts; 2 lower washers
Tilt horn retainer	2 x M2 x 8 COUNTERSUNK	2 M2 nuts; 2 rear washers
Camera / hood	4 x M2 x 12	4 M2 nuts; 8 M2 washers
Pivot support	2 x M2 x 12	2 M2 nuts; 4 M2 washers
Tilt pivot	1 x M3 x 12	1 M3 nut; 0.5 mm washer + 0.7 mm shim
Electronics cover	4 x M3 x 35	4 M3 nuts; 8 M3 washers
Servo centres	2 x original horn screws	Thread and length depend on servo

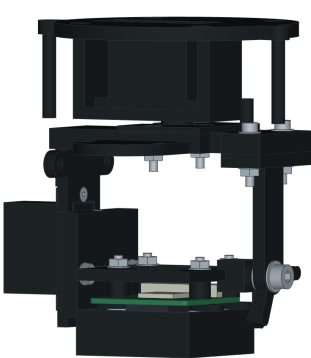
Also: two converter ties, one capacitor tie, and ties/soft sleeves for the incoming wire and CSI anchor pads. Do not use a cable tie as the independent tensile termination of a positioning line.

12 What the CAD check proves

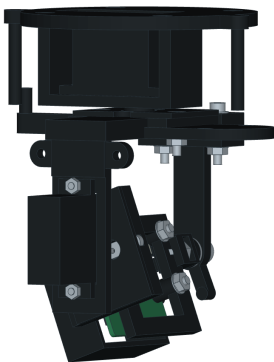
Checked against the declared nominal hardware. Real parts, print shrinkage, bolt tolerances and cable behaviour still need inspection.



PAN -90 / TILT 70



PAN 0 / TILT 0



PAN +90 / TILT 70

Check	Result / limit
Rigid travel grid	555 poses; 5-degree steps; no intersections above 0.005 mm3
Mechanical stops	Contact beyond +/-95 pan and -5 / +75 tilt; usable targets +/-90 and 0..70
Assembly and service	Insertion, cover-removal and short-driver paths checked in the supplied report
Optical opening	Nominal camera viewing volume checked against represented solids
Print mass	163.3 g at solid PETG density 1.27 g/cm3, BEFORE electronics and metal hardware

The original 170 g flying ceiling is not demonstrated by this bench design. Slicer infill changes mass; weigh the complete build. A lighter chassis/cover and a new mass check are needed before suspended use.

13 Final bench inspection

Support the spider on a fixture so the camera hangs free. Do not rest the payload on its optical hood or force a geared servo by hand.

1 Inspect the retained parts

No loose PCB, horn, pivot nut or strap. Countersunk heads fully seated. Camera centre screws installed. The pivot remains free and has no excessive side play.

2 Check cables before powered motion

Verify both slack loops and all four travel corners with the drive disengaged where practical. Calibrate narrow software limits first, then increase only after observing clearance.

3 Check power and imaging

Verify the converter output before attaching the Pi. Check for brownouts during servo motion, focus and capture. Keep hands, loose wires and tools away from the moving brackets.

4 Record what was built

Measure servo/converter dimensions, revisions, print settings, complete mass, movement limits and faults. No load, fatigue, weather or suspended-operation test has been performed here.

Pack contents

41 checked STLs, assembled GLB, current source snapshot, assembly/figure transforms, source hashes and JSON check reports. Only models/printable contains fabrication parts; the fit coupon is not installed.

Sources and open interfaces

Full source URLs and assumptions: sources.json. Line terminations, docking-stud interface, actual ribbon behaviour, electrical performance and flying mass remain open; they are not supplied as verified interfaces.